

Qin Shi

✉ shi622@purdue.edu • 🏠 shqii1j.github.io

Education

Purdue University

Doctor of Philosophy in Statistics

Indiana, IN

2023–2027

National University of Singapore

Master of Statistics

Singapore

2022–2023

Shanghai University of Finance and Economics

Bachelor of Science in Data Science

Shanghai, China

2018–2022

Publications

- [1] **Qin Shi***, Amber Yijia Zheng*, Qifan Song, and Raymond A. Yeh. Knowledge distillation detection for open-weights models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [2] Jiawei Du, **Qin Shi**, and Joey Tianyi Zhou. Sequential subset matching for dataset distillation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Research & Work Experience

Optum AI

Research Intern

Remote

May 2026 – Aug 2026

- Research on generative models for structured data extraction from charts.

Purdue University — Department of Statistics

Research Assistant

West Lafayette, IN

July 2024 – May 2026

Advisor: Prof. Qifan Song

- Built acceleration techniques for **diffusion language models** to improve inference speed and token-level consistency.
- Developed methods to detect knowledge distillation in image classification, **text-to-image**, and **large language models**, improving transparency and reliability of open-weight models.
- Designed data distillation algorithms for **vision-language datasets**, achieving better efficiency in multimodal tasks.

A*STAR's Institute of High-Performance Computing

Research Intern

Singapore

Sept. 2022 – June 2023

Mentors: Prof. Joey Tianyi Zhou, Dr. Jiawei Du

- Developed a generalized training framework for **data distillation (DD)** that significantly improved the performance of gradient-matching-based methods.
- Resolved the coupling issue in synthetic datasets, enabling better representation of high-level semantic features.
- Achieved state-of-the-art results across multiple benchmarks, and the approach has been adopted for follow-up research on efficient model training.

YumChina

Data Scientist Intern

Shanghai, China

Apr. 2022 – June 2022

- Built a large-scale **multi-objective recommendation system** that was successfully deployed in production, achieving an AUC of 0.73 on offline evaluation.
- Designed a deep structured semantic model to generate meal-combination templates based on user purchasing behavior, improving personalization and sales insights.

MarcPoint

NLP Algorithm Engineer Intern

Shanghai, China

Sept. 2021 – Jan. 2022

- Designed and deployed a text-filtering algorithm for social media data, achieving an 80% precision rate on human-labeled samples with processing speed of 4 minutes per million entries.
- Implemented an ACNN-CRF model for named entity recognition (NER), improving runtime efficiency by 11% compared to BERT-CRF while maintaining accuracy.
- Contributed to the MEMO project on NER evaluation, proposing new dataset partitioning and evaluation methods for entity recognition models.

Academic Services

Conference Reviewer: European Conference on Computer Vision (ECCV), Neural Information Processing Systems (NeurIPS), International Conference on Computer Vision (ICCV)